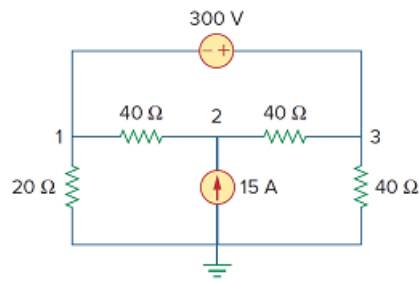


Problem

Determine the node voltages in the circuit in Fig. 3.67 using nodal analysis.

Figure 3.67:



Step-by-step solution

Step 1 of 6

Refer to Figure 3.67 in the text book.

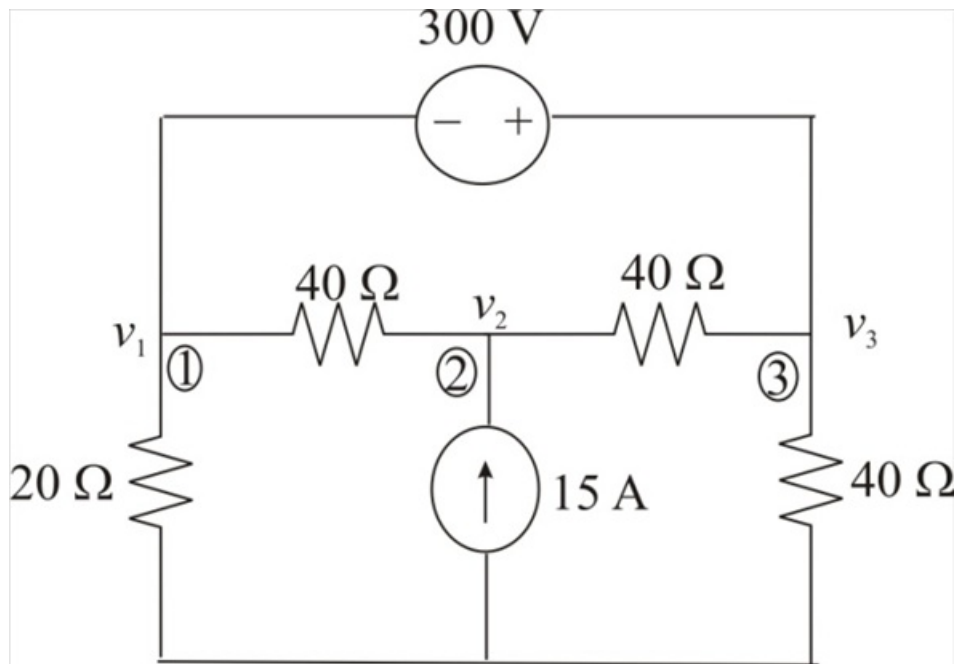


Figure 1

Step 2 of 6

Notice that the super node analysis can be applied at nodes 1 and 3.

Consider the relation between the voltages v_1 and v_3 .

$$v_3 - v_1 = 300$$

$$v_3 = v_1 + 300 \dots\dots (1)$$

Therefore, the expression for the voltage v_3 is, $v_1 + 300$.

Step 3 of 6

Apply super node analysis at nodes 1 and 3.

$$\frac{v_1}{20} + \frac{v_1 - v_2}{40} + \frac{v_3}{40} + \frac{v_3 - v_2}{40} = 0$$

$$\frac{2v_1 + v_1 - v_2 + v_3 + v_3 - v_2}{40} = 0$$

$$2v_1 + v_1 - v_2 + v_3 + v_3 - v_2 = 0$$

$$3v_1 - 2v_2 + 2v_3 = 0$$

Substitute $v_1 + 300$ for v_3 .

$$3v_1 - 2v_2 + 2(v_1 + 300) = 0$$

$$3v_1 - 2v_2 + 2v_1 + 600 = 0$$

$$5v_1 - 2v_2 + 600 = 0$$

$$2v_2 = 5v_1 + 600$$

$$v_2 = \frac{5v_1 + 600}{2}$$

$$v_2 = 2.5v_1 + 300 \text{ (2)}$$

Therefore, the expression for the voltage v_2 is $2.5v_1 + 300$.

Step 4 of 6

Consider the nodal equation at node-2.

$$\frac{v_2 - v_1}{40} + \frac{v_2 - v_3}{40} - 15 = 0$$

$$\frac{v_2 - v_1 + v_2 - v_3 - 600}{40} = 0$$

$$v_2 - v_1 + v_2 - v_3 - 600 = 0$$

$$2v_2 - v_1 - v_3 - 600 = 0$$

Substitute $2.5v_1 + 300$ for v_2 and $v_1 + 300$ for v_3 .

$$2(2.5v_1 + 300) - v_1 - (v_1 + 300) - 600 = 0$$

$$5v_1 + 600 - v_1 - v_1 - 300 - 600 = 0$$

$$3v_1 - 300 = 0$$

$$3v_1 = 300$$

$$v_1 = \frac{300}{3}$$

$$= 100 \text{ V}$$

Therefore, the value of the voltage v_1 is, 100 V.

[Comments \(1\)](#)



Anonymous

Why is it minus 15?

Step 5 of 6

Recall equation (1).

$$v_3 = v_1 + 300$$

Substitute 100 V for v_1 .

$$\begin{aligned} v_3 &= 100 + 300 \\ &= 400 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_3 is, 400 V.

Step 6 of 6

Recall equation (2).

$$v_2 = 2.5v_1 + 300$$

Substitute 100 A for v_1 .

$$\begin{aligned} v_2 &= 2.5(100) + 300 \\ &= 250 + 300 \\ &= 550 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_2 is, 550 V.